CombiMOTS: Combinatorial Multi-Objective Tree Search for Dual-Target Molecule Generation

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Abstract

Dual-target molecule generation, which focuses on discovering compounds capable of interacting with two target proteins, has garnered significant attention due to its potential for improving therapeutic efficiency, safety and resistance mitigation. Existing approaches face two critical challenges. First, by simplifying the complex dual-target optimization problem to scalarized combinations of individual objectives, they fail to capture important trade-offs between target engagement and molecular properties. Second, they typically do not integrate synthetic planning into the generative process. This highlights a need for more appropriate objective function design and synthesis-aware methodologies tailored to the dual-target molecule generation task. In this work, we propose CombiMOTS, a Pareto Monte Carlo Tree Search (PMCTS) framework that generates dual-target molecules. CombiMOTS is designed to explore a synthesizable fragment space while employing vectorized optimization constraints to encapsulate target affinity and physicochemical properties. Extensive experiments on real-world databases demonstrate that CombiMOTS produces novel dual-target molecules with high docking scores, enhanced diversity, and balanced pharmacological characteristics, showcasing its potential as a powerful tool for dual-target drug discovery. The code and data is accessible through `https://github.com/Tibogoss/CombiMOTS`.

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1. Introduction

Complex diseases and disorders such as cancers are generally caused by intricate interacting pathways. To treat them, traditional therapies focused on the single-target paradigm by developing drugs selective towards a unique mechanism, therefore "blind to other processes" yet equally involved in biological systems (Medina-Franco et al., 2013). Single-target drugs are prone to resistance and can risk the activation of unwanted compensatory signaling pathways (Yang et al., 2024b). To overcome this limitation, a rising field of study aims to develop treatments which exhibit better efficacy by interacting with multiple related targets. An intuitive approach known as combination therapy (Kummar et al., 2010; Foucquier & Guedj, 2015) consists in administering various synergetic single-target medications to address distinct but related pathogenic mechanisms. However, these approaches often risk adverse effects or poor treatment adherence (Ye et al., 2023). As promising alternatives, dual-target drugs are increasingly drawing attention, demonstrated by a trend in FDA-approved drugs (Li et al., 2016; Lin et al., 2017; Layman et al., 2024). They involve one agent simultaneously interacting with two targets with better pharmacokinetics and safety profiles, thus avoiding undesirable drug-drug interactions (Hopkins et al., 2006). This inevitably imposes new challenges: (1) The Structure-Activity Relationship (SAR) profile of considered compounds must consider different biological targets (Morphy & Harris, 2012; Raghavendra et al., 2018). (2) The identified compounds have to be synthesizable through known experimental protocols.

In the past few years, deep generative models (Zeng et al., 2022) and geometric deep learning (Powers et al., 2023) tackled SAR for single-target drug design, but the data scarcity and expensive computational cost limit their application on the dual-target setting. Recently, machine learning models allowed to bridge the gap between predictive and intuitive dual-target drug design (Feldmann et al., 2021). Notably, great efforts were made in Fragment-Based Drug Discovery (FBDD) by leveraging property predictors and oracles to extract, combine and autoregressively complete molecular substructures, supposedly responsible for the bioactivity of known compounds on both given targets

(Jin et al., 2020; Lee et al., 2023; Chen et al., 2024). Nevertheless, the key limitation of such methods is the translation from a multi-objective problem to a single-objective formulation. Despite integrating various constraints, their reward function aggregates weighted objectives into a scalar value without considering potential conflicts, leading to overall high-scoring generations with imbalanced properties (Fromer & Coley, 2023; Luukkonen et al., 2023). More specifically, by assuming a convex search space, distinguishing between competing characteristics becomes intractable and resulting molecules can exhibit idealistic properties while being nearly impossible to synthesize (Gao & Coley, 2020; Stanley & Segler, 2023). Furthermore, solely relying on desirability metrics during the generative process cannot guarantee true synthesizability (Vinkers et al., 2003; Ertl & Schuffenhauer, 2009). Consequently, new frameworks also need to condition their processes on synthetic routes (Segler & Waller, 2017; Segler et al., 2018). Swanson et al. (2024) applied Monte-Carlo Tree Search (MCTS) to assemble make-on-demand molecules with associated synthetic procedures yielding an average success rate over 80%, but fall again into the single-objective limitation.

Other fields such as robotics and the computational game domain employ Pareto optimization to explore attainable solutions presenting optimal tradeoffs regardless of competing objectives and property imbalance (Luc, 2008). This flexibility enables to adjust priorities and selection criteria as new information becomes available, without needing to restart the optimization process with different weightings. Few recent works apply Pareto optimality to MCTS in the fields of multi-constrained retrosynthesis planning (Lai et al., 2025) and atom-by-atom molecule SMILES generation (Yang et al., 2024a). However, its application to dual-target molecule generation remains underexplored as the choice of objectives, state space or action space is challenging.

In this paper, we draw inspiration from Pareto MCTS (PMCTS) to propose a novel approach to combinatorial multi-objective drug design. We propose a fragment-based molecule generation approach that optimizes multi-constrained properties of dual-targets within a synthesizable space. To achieve this: (1) We perform search space reduction by extracting target-aware fragments from known target inhibitors and mapping them to industry-available building blocks. (2) We extend PMCTS to a combinatorial framework alongside vectorized objectives tailored for dual-target molecule generation, ultimately generating interpretable compounds exhibiting better tradeoffs in all objectives compared to competing methods. We experimentally validate the effectiveness of our method through the case of dual inhibitor generation as a real world scenario. We summarize our contributions as follows:

- We propose Combinatorial Multi-Objective Tree Search (CombiMOTS), extending MCTS to Pareto optimization and fragment-based drug discovery to narrow down a large molecule fragment space into a synthesizable one, specific to any target pair.
- Through the task of dual inhibitor generation, we demonstrate and analyze CombiMOTS' ability to find optimal candidates across conflicting objectives in complex multi-objective multi-target settings.
- In particular, we evaluate our model on the GSK3β-JNK3 pair and curate data for two new target pairs, EGFR-MET and PIK3CA-mTOR to illustrate the utility of CombiMOTS in dual-target drug design.
- On all experiments, CombiMOTS consistently generates molecules that yield better diversity and trade-off between docking score, drug-likeness, and synthetic accessibility.

2. Related Work

Multi-Objective Molecule Generation The success of prior generative models for single-target/objective drug design does not trivially apply to multi-objective parameterization (Angelo et al., 2023). These methods commonly optimize embeddings of molecular sequences or graphs, relying on bayesian inference or reinforcement learning. Multi-objective molecule generation approaches can be categorized into three categories. The first category uses encoder-decoder architectures (Jin et al., 2018; 2020) or autoregressive frameworks (Gong et al., 2024; Yang et al., 2024a) to attempt learning continuous latent representations of target compounds to generate candidates with encouraging predicted properties. However, they highly depend on the quality of the learnt latent space, which is burdensome in multi-objective tasks where data is scarce. The second category being reinforcement learning, operates in the explicit chemical space to allow more robust training, but are hard to train due to reward sparsity when combining multiple objectives (Guimaraes et al., 2017; Olivecrona et al., 2017; Popova et al., 2018). More specific to dual-target drug design, Li et al. (2018); Jin et al. (2020); Xie et al. (2021) utilize a machine learning property predictor to guide the generation of bio-active compounds, but still require high-quality labeled data to properly work. As the third category, Structure-Based Drug Design (SBDD) methods integrate binding affinity indicators to moderate data scarcity while enhancing SAR with target proteins (Chen et al., 2024; Zhou et al., 2024). Compared to these methods, we utilize both property predictors and binding score oracles to cooperatively account for pharmacological characteristics and dual-target affinity.

Fragment-Based Molecule Generation Fragment-based approaches have emerged as a promising direction in drug discovery. The intuition is to derive multi-property candidates from initial single-property molecular substructures as building blocks. Recent research in this area can be broadly categorized into fragment extraction and assembly methods for generation. For fragment extraction, early approaches relied on rule-based bond breaking selection (Yang et al., 2021) or motif frequency-based selection (Kong et al., 2022). The drawback of these heuristic methods generally is to not consider target molecular properties. This motivated succeeding works to try identifying property-aware substructures. Jin et al. (2020); Chen et al. (2024) extract and assemble core fragments from full molecules rather than true fragments, leading to limited novelty and diversity in generation. For generation methods, MCMC sampling strategy can be applied for its flexibility in adding and deleting fragments (Xie et al., 2021), while reinforcement learning (Tan et al., 2022; Du et al., 2023), and VAE-based models for fragment assembly (Maziarz et al., 2021) remain widely used. Interestingly, Swanson et al. (2024) proposed a method that does not require to extract fragments as building blocks. They use the industry-ready Enamine REadily AccessibLe (REAL) Space database (Grygorenko et al., 2020), with building blocks and reaction templates as inputs to MCTS, with a single property predictor guiding the space navigation modeled as chemical reactions. This presents the advantage of keeping a high chance to synthesize identified leads in real world laboratory settings. We propose to extend their work to the dual-target molecule generation task by leveraging property-aware available building blocks, subsequently assembled using a vectorized multi-objective parametrization guided by oracles tailored to the target proteins.

3. Preliminary - Pareto Optimization

Multi-objective problems involve optimizing several objectives, often competing. A common approach is to aggregate them into one scalarized reward, naturally inducing oversimplification of the problem's complexity by depending on importance weighting and its *a priori* knowledge. Pareto optimization (El-Sharkawi, 2005) alleviates this issue by maintaining a vectorized problem formulation, thus defining sets of solutions whose components are all considered equal, allowing proper optimization in the attainable solution space. Without loss of generality, we assume a maximization problem over $D \in \mathbb{R}$ objectives, where each solution is characterized by a D-dimensional property vector.

Definition 3.1 (Pareto Dominance)**.** Let $\mathbf{X}$ and $\mathbf{Y}$ be two distinct objects, each associated with their respective objective vectors $\mathbf{P}_X, \mathbf{P}_Y \in \mathbb{R}^D$, where $P_{X,d}$ and $P_{Y,d}$ denote the values of X and Y on the d-th objective, respectively. We define Pareto dominance as follows: X dominates Y, denoted by $X \succ Y$ (or equivalently $Y \prec X$), if and only if:

i) No coordinate of $\mathbf{P}_X$ is smaller than the corresponding element of $\mathbf{P}_Y$:

$$\forall d \in \{1, 2, \ldots, D\}, P_{X,d} \geq P_{Y,d} \ . \tag{1}$$

ii) At least one coordinate of $\mathbf{P}_X$ is larger than the corresponding element in $\mathbf{P}_Y$:

$$\exists d \in \{1, 2, \ldots, D\} \text{ s.t. } P_{X,d} > P_{Y,d} \ . \tag{2}$$

If only the first condition is satisfied, X is said to weakly-dominate Y, denoted by $X \succeq Y$ or $Y \preceq X$. When neither $X \succeq Y$ nor $Y \succeq X$ holds, X and Y are said incomparable $(X \| Y)$.

Remark 3.2. i) and ii) can be summarized as: $X \succ Y$ if X is "never worse" and "at least better than Y in one objective".

Definition 3.3 (Pareto Fronts)**.** Given an non-empty set of vectors $\mathcal{S} \subset \mathbb{R}^D$ obtained from candidate solutions found in the objective space, the first Pareto front (or Pareto optimal set) consists of all non-dominated solutions:

$$\mathcal{S}_1 = \{\mathbf{P}_X \in \mathcal{S} : \nexists \mathbf{P}_Y \in \mathcal{S} \text{ s.t. } Y \succ X\} \ . \tag{3}$$

Subsequent Pareto fronts $\mathcal{S}_k$ are defined recursively by excluding all solutions belonging to the preceding $\{S_i\}_{i=1}^{k-1}$:

$$\mathcal{S}_k = \{\mathbf{P}_X \in \mathcal{S} : \nexists \mathbf{P}_Y \in \mathcal{S} \setminus \bigcup_{i=1}^{k-1} \mathcal{S}_i \text{ s.t. } Y \succ X\} \ . \tag{4}$$

Remark 3.4. All solutions from the same Pareto front are considered equivalent, as they are either identical or incomparable:

$$\forall \mathcal{S}_k \subset \mathcal{S}, \forall (X, Y) \in \mathcal{S}_k, (X \| Y) \ . \tag{5}$$

4. Proposed Approach: CombiMOTS

We propose CombiMOTS, a PMCTS method designed to navigate a dual-target specific synthesizable space by leveraging the strengths of both FBDD and Pareto optimization. We first outline in Sec.4.1 how to reduce a large chemical space to available dual-target-specific building blocks. We then describe our reaction-based PMCTS implementation in Sec.4.2. We provide additional implementation details in Appendix D.3.

4.1. Synthesizable Search Space Reduction

The estimated size if the drug-like chemical space exceeds 10^{33} molecules, making exhaustive search for compounds with desirable chemical and structural properties computationally infeasible (Polishchuk et al., 2013).

To address this challenge, we narrow the search space using curated subsets of Enamine REAL Space (Grygorenko et al., 2020), refined by Swanson et al. (2024). This ensures that our method remains computationally tractable while maintaining its practical relevance. Given a target pair, we reduce the search space by (1) identifying target-aware fragments from known active inhibitors, and (2) aligning them with the REAL Space building blocks and reactions (Figure 1).

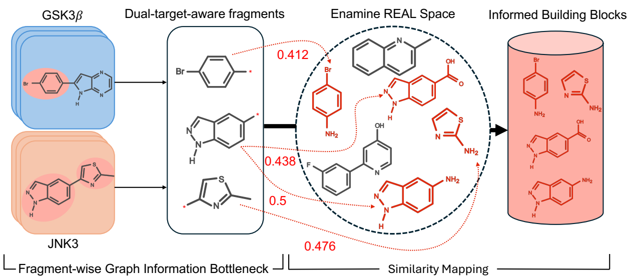

Figure 1. Search Space Reduction. Property-aware fragments are extracted for each target using Fragment-wise Graph Information Bottleneck (Lee et al., 2023). By applying a Tanimoto similarity threshold to a synthesizable search space, we curate a final set of informed industry-ready building blocks.

Fragment Extraction To extract chemically meaningful fragments from active molecules, we employ Fragment-wise Graph Information Bottleneck (FGIB) (Lee et al., 2023). FGIB provides a sophisticated approach to extracting substructures using BRICS decomposition (Degen et al., 2008) from known active compounds, highly contributing to their target property. Its theoretical foundation lies in the Graph Information Bottleneck (GIB) principle (Wu et al., 2020), where the objective function balances between preserving property-predictive information and compressing an original molecular graph representation. This is achieved through a dual optimization process, where the goal is to simultaneously maximize information between extracted fragments and target properties, while minimizing the mutual information between fragments and the original graph. Given a molecule graph G and its target property Y, GIB identifies an informative set of fragments G^{sub} by optimizing:

$$\min_{G^{\text{sub}}} -I(G^{\text{sub}}, Y) + \beta I(G^{\text{sub}}, G) \,. \tag{6}$$

where $I(.|.)$ denotes mutual information and β is a positive hyperparameter balancing informativeness and compression.

FGIB utilizes Message Passing Neural Networks (MPNN) to compute graph embeddings and introduces a novel noise injection mechanism that modulates information flow based on fragment importance. This mechanism enhances our model's ability to identify chemically meaningful substructures by selectively preserving informative signals, while reducing redundancy.

Search Space Reduction To ensure interpretable and realistic generations, the core implementation of CombiMOTS relies on predefined building blocks from the Enamine REAL Space. After obtaining dual-target-aware fragments, we curate the final set of building blocks by mapping learnt fragments to existing Enamine building blocks with a Tanimoto similarity above a fixed threshold (Figure 1). We empirically find in Section 5.7.2 that using ECFP4 Morgan fingerprints and a threshold value of 0.4 generally leads to an appropriate search space size ($\sim$14k blocks and $\sim$25M possible reaction products).

4.2. Pareto Monte-Carlo Tree Search

4.2.1. Monte-Carlo Tree Search

MCTS is a decision-making algorithm that systematically explores a search space by balancing exploration and exploitation through four distinct phases. (i) The **selection** phase aims at traversing a search tree from the root node to a leaf node, using the Upper Confidence Bound (UCB) formula to greedily select the most promising actions based on both their estimated UCB value and the uncertainty of those estimates. Given a synthesis tree T to search, a node n with parent p, we denote C a positive exploration constant, $R(n)$ the cumulative reward of n and $N(n)$ the number of times n was visited:

$$\text{UCB}(n) = \underbrace{\frac{R(n)}{N(n)}}_{\text{exploitation}} + C \underbrace{\sqrt{\frac{\ln(N(p))}{N(n)}}}_{\text{exploration}} \,. \tag{7}$$

The 'exploitation' term favors selecting high-property nodes, while the 'exploration' term encourages selecting less explored nodes, jointly guiding the traversal of both promising and uninvestigated outcomes. (ii) In the **expansion** phase, reaching a leaf node triggers the creation of child nodes, each representing a new potential state or action in the search space. (iii) The **simulation** (rollout) phase iteratively applies steps (i) and (ii) from the newly expanded node until reaching a terminal state or predefined depth. (iv) In the **backpropagation** phase, the statistics of all nodes along the traversed path are updated, propagating simulation outcomes upward to refine value estimates and visit counts. This iterative process continues until computational constraints or a convergence criterion are met, progressively constructing a search tree that balances exploration and exploitation while ensuring sufficient search space coverage.

4.2.2. Pareto MCTS for Dual-Target Molecules

We propose an extended version of MCTS that incorporates Pareto optimization through vectorized properties, integrating property predictors and docking oracles based on combinatorial chemistry principles (Liu et al., 2017) to identify synthetically accessible compounds that jointly target two proteins.

Algorithm 1 High-Level CombiMOTS Algorithm

```
Input: Reactions R, rollouts n_rollout
Require: Multi-objective Oracles
Initialize MCTS tree T with empty root node
for i = 1 to n_rollout do
    v ← Rollout(T.root)
end for
return Pareto optimal molecules from visited nodes

function Rollout(node)
    if node is a product then
        return Oracles(node) ▷ Backpropagation
    end if
    children ← GetChildNodes(node) ▷ Expansion
    selected ← ParetoSelection(children)
    v ← Rollout(selected)
    if selected is a product then
        v ← Elem-wise-max(v, selected.P) ▷ Backprop.
    end if
    Update statistics of selected
    return v
end function
```

The major modification from the single-objective version MCTS described in Section 4.2.1 is the design of multi-dimensional property vectors, enabling element-wise optimization. This formulation allows the application of Pareto principles (Section 3) by optimizing vector coordinates across the full objective space, but it also requires modifications of the MCTS core steps. In this setting, tree traversal can no longer greedily select the highest-scoring node. Given the property vector $\vec{Ora}(n)$ obtained from $Oracles$ and D the number of objectives, Equation (7) becomes the Pareto UCB formula (PUCB):

$$\overrightarrow{PUCB}(n) = \frac{\vec{R}(n)}{N(n)} + C \times \overrightarrow{Ora(n)} \sqrt{\frac{\ln(D) + 4 \times \ln(1 + N(p))}{1 + N(n)}}. \tag{8}$$

The Figure 2 and Algorithm 1 illustrate a complete rollout of CombiMOTS, while Algorithm 2 provides a detailed description of the algorithm. During the selection step, after computing the PUCB vector of all child nodes, the non-dominated nodes form a local Pareto front, from which the next selected node is randomly sampled. In our problem, each node represents a tuple of multiple building blocks. Upon reaching a leaf node, the expansion step generates new child nodes in two ways: (a) by applying all possible reactions to the building blocks in the node and creating one node per product, or (b) incorporating all matching reactants of each building block and creating one node per addition. The rollout iteratively repeats these steps until a reaction product is reached (terminal node). When a terminal node is reached, the backpropagation step updates the reward vectors of all nodes along the traversal path, before initiating the next rollout. The algorithm ends once the specified number of rollouts is completed, returning all encountered reaction products.

5. Experiments

5.1. Datasets

As our work focuses on dual-target molecule generation, we demonstrate the practical utility of CombiMOTS in real-world scenarios by evaluating it on three disease-related protein target-pairs. Notably, we curate and release new datasets for the EGFR-MET and PIK3CA-mTOR pairs. Detailed data statistics and curation methods are provided in Appendices D.1 and M.

GSK3β-JNK3 These are two kinases related to Alzeihmer's Disease (AD) (McCubrey et al., 2014; Koch et al., 2015). Designing dual-inhibitors for these targets open possibilities towards more efficient AD therapies. We utilize the data curated by Li et al. (2018) as a commonly used benchmark for dual-target generation (Jin et al., 2020; Xie et al., 2021; Chen et al., 2024).

EGFR-MET In non-small cell lung cancer (NSCLC), resistance to EGFR inhibitors often involves compensatory activation of MET, leading to tumor progression (Fang et al., 2024). Therefore, a dual-target molecule that simultaneously inhibits EGFR and MET could overcome drug resistance and improve therapeutic outcomes (Ren et al., 2024). This strategy is supported by multiple studies showing the potential benefits of combined EGFR and MET inhibition in patients with EGFR mutations and MET amplification or overexpression (Fang et al., 2024; Ren et al., 2024).

PIK3CA-mTOR Mutations in PI3K, particularly PIK3CA, and dysregulation of mTOR contribute to cancer cell growth, proliferation, and survival (Fruman & Rommel, 2014). Dual inhibitors targeting both PI3K and mTOR provide a more comprehensive blockade of this pathway, addressing potential resistance mechanisms, as demonstrated by the PI3K/mTOR dual inhibitor VS-5584 (Hart et al., 2013). Simultaneous inhibition of both targets is expected to more effectively suppress PI3K-mTOR signaling compared to isoform-selective inhibitors, potentially overcoming feedback activation that limits single-target therapies (Rodrik-Outmezguine et al., 2016).

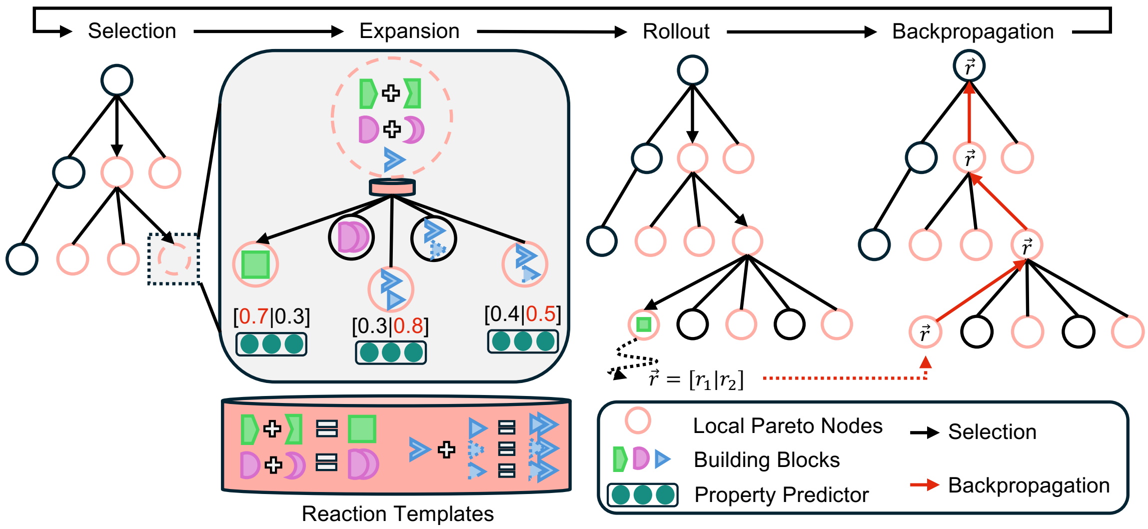

Figure 2. One full iteration of the CombiMOTS Algorithm. (i) Selection from local Pareto fronts occurs until a leaf node to expand is found. (ii) During expansion, two types of nodes are created by consulting the reaction templates: those representing products from current building blocks and those incorporating compatible reactants. Upon creation, oracles predict properties for all child nodes to establish their local Pareto front. We illustrate property vectors with two objectives. (iii) Selection and Expansion steps iteratively occur until a reaction product is found. (iv) The property vector of the reaction product is backpropagated up the path.

5.2. Baselines

RationaleRL (Jin et al., 2020) identifies important structural components from active compounds through MCTS. It then systematically merges these components while maintaining their shared structural elements, followed by targeted fine-tuning of a generative model to append appropriate side chains. **REINVENT** (v3.2) (Blaschke et al., 2020) adopts a sequence-based approach, utilizing a recurrent neural network architecture to generate molecular SMILES strings. The framework incorporates reinforcement learning techniques to optimize multiple molecular properties simultaneously, enabling the design of compounds with activity across multiple objectives. **MARS** (Xie et al., 2021) selects the top 1,000 fragments from the ChEMBL dataset based on their size and frequency. The obtained fragments have less than 10 heavy atoms with frequent occurrence in the dataset. It employs graph neural networks (GNNs) and Markov Chain Monte Carlo (MCMC) sampling to refine molecules with enhanced properties, from addition or deletion of these fragments.

5.3. Evaluation Metrics

For each model, we generate N = 10,000 molecules for all target pairs, then compare the performance of models to assess their ability to generate original and biochemically potent compounds. We evaluate generations with metrics fitting two categories:

Originality We gauge the models' ability to generate original compounds using the following metrics. **Validity (%)**: The proportion of molecules adhering to all chemical rules. **Uniqueness (%)**: The proportion of distinct molecules within the generated set, where a low value suggests repetitive generation. **Novelty (%)**: As defined by Olivecrona et al. (2017), the proportion of generated molecules whose nearest training set dual-active neighbor has a Tanimoto similarity score below 0.4, reflecting the model's capacity to generate sufficiently different, yet desirable compounds. **Diversity (%)**: The proportion of generated molecules with a pairwise Tanimoto similarity below 0.4 across all generations, indicating the model's ability to explore broad regions of the chemical space rather than a narrow subset. **Activity Success Rate (%)**: The proportion of generated molecules predicted to be active towards both target proteins, indicating the model's ability to identify potential dual-inhibitors. We denote such molecules as "dual-actives".

Quality We interpret property distributions by visualizing density plots of dual-active compounds. We analyze: **Docking Score** (DS) as an approximation of binding affinity between target proteins and molecule ligands. A lower docking score indicates better molecular interactions. **Quantitative Estimate of Drug-likeness** (QED) quantifies the potential of a compound presenting desirable drug capabilities based on properties such as topological polar surface area and the number of hydrogen donors/acceptors. Ranging from 0 to 1, a higher score is preferable. **Synthetic Accessibility score** (SA) measures how difficult a given compound is to synthesize, based on fragment contribution and complexity penalty. Ranging from 1 to 10, compounds with a low SA score are easier to obtain.

Details on used oracles and molecular docking settings are given in Appendix D.3.

5.4. Objectives

RationaleRL and MARS are reproduced in their best reported settings. Following Li et al. (2018), they use random forest (RF) classifiers for each target, alongside QED and SA scores to condition their generation. REINVENT is reproduced with the same property predictors with QED score only as SA score is not supported.

The objectives of CombiMOTS integrate both biological activity and structural constraints. Following Li et al. (2018); Jin et al. (2020); Xie et al. (2021); Swanson et al. (2024), we use activity towards both targets as an empirical estimator of inhibition, derived from real experimental assays used to train machine learning predictors. For instance, the GSK3β-JNK3 training data originates from PubChem and ChEMBL (Gaulton et al., 2012; Kim et al., 2016), filtered based on bioactivity data, particularly IC_{50} values. We employ a Chemprop (D-MPNN) classifier for multi-task prediction of dual-target activity. Unlike baselines, we prioritize molecular docking scores for both targets over QED and SA. Specifically for dual-inhibitor design, we argue that to accurately consider SAR during generation, "We Should at Least Be Able to Design Molecules That Dock Well" (Cieplinski et al., 2020): CombiMOTS successfully captures ligand-binding affinity, as showcased in Section 6.1. Details and justification on objective design are given in Appendices D.3, E.1.2 and E.1.3.

We conduct additional experiments in Appendix E. They include comparative analysis of: (i) A scalarized version of CombiMOTS to emphasize the utility of Pareto MCTS. (ii) A variant prioritizing QED/SA over docking scores and a six objective version which also incorporates QED and SA scores. (iii) Implementations of MARS and REINVENT aligned with our objectives by replacing QED and SA with docking score. (iv) A comparison against AIxFuse (Chen et al., 2024), a model using collaborative reinforcement learning and active learning, on the DHODH-RORγt pair to investigate how our model handles data scarcity.

5.5. Experimental Settings

For RationaleRL, REINVENT and MARS, we generate N = 10,000 molecules across all three target pairs. As CombiMOTS is fundamentally a search method, it does not 'generate' a fixed number of outputs but instead returns all reaction products within a computational budget. We fix $n_{rollout}$ = 50,000 for all tasks (GSK3β-JNK3, EGFR-MET, PIK3CA-mTOR) and randomly sample 10,000 found dual-active molecules. *Therefore, we do not need to compute activity success rate for CombiMOTS.*

5.6. Results

As shown in Figures 3 and 7, CombiMOTS consistently identifies optimal tradeoffs across all three target pairs under each metric. In contrast, competing methods exhibit inconsistent performance across target pairs, often generating compounds with suboptimal properties or excelling in a single objective while underperforming in others. Notably, displayed density distributions are normalized, highlighting the rarity of high-quality candidates produced by alternative methods. Furthermore, Table 1 suggests that REINVENT and RationaleRL can generate invalid SMILES, and all models struggle to generate unique, novel and diverse compounds on all tasks. In every setting, CombiMOTS outperforms other methods in generating novel and diverse molecules with well-balanced physicochemical and structural properties, as visualized in Figure 4. Finally, we compute the respective first Pareto front of all models and report in Table 2 their average R2-distance to the utopia point, as well as the size of their optimal set (details in Appendix D.3): our approach steadily finds more Pareto optimal solutions. This evidence demonstrates the effectiveness of our approach in navigating the complex multi-objective landscape of dual-target drug design.

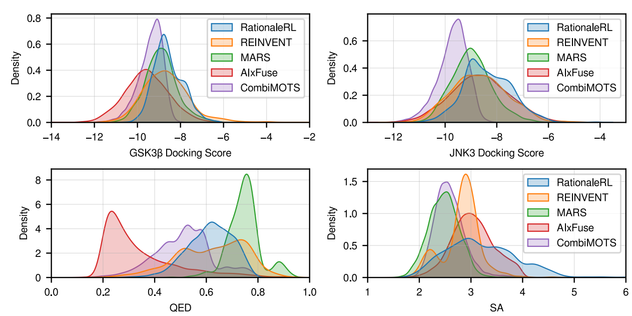

Figure 3. Normalized distributions of the generated molecules across dual-docking scores, QED and SA scores on the GSK3β-JNK3 target pair.

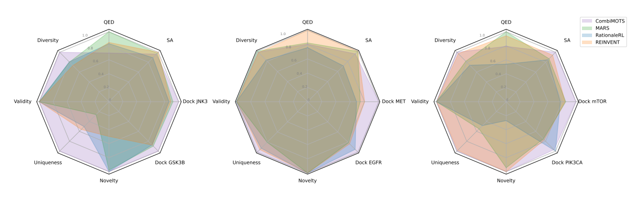

Figure 4. Radar charts summarizing performance on the GSK3β-JNK3 (left), EGFR-MET (middle) and PIK3CA-mTOR (right) tasks. We report average values of originality metrics and median values of quality metrics, normalized to the best score on each axis.

5.7. Ablation Studies

To demonstrate the effectiveness of the Search Space Reduction step, we conduct experiments to emphasize the role of FGIB in fragment extraction (Section 5.7.1) and the impact

Table 1. Performance comparison across different target pairs and molecular generation models based on 10,000 sampled molecules from three runs. *(**: $p < 0.001$; *: $p < 0.01$)*

Target Pairs	Models	Metrics				
		Valid. (%)	Uniq. (%)	Novel. (%)	Div. (%)	Act. SR (%)
GSK3β-JNK3	RationaleRL	99.95$_{\pm 0.01}$	50.46$_{\pm 0.05}$	**100**$_{\pm 0.00}$	70.92$^{**}_{\pm 0.01}$	**99.97**$_{\pm 0.01}$
	REINVENT	99.86$_{\pm 0.06}$	56.14$_{\pm 0.77}$	49.72$_{\pm 0.37}$	64.52$^{**}_{\pm 0.19}$	99.33$_{\pm 0.15}$
	MARS	**100**$_{\pm 0.00}$	26.02$_{\pm 3.04}$	97.87$_{\pm 0.56}$	70.64$_{\pm 2.85}$	99.86$_{\pm 0.10}$
	CombiMOTS	**100**$_{\pm 0.00}$	**100**$_{\pm 0.00}$	99.96$_{\pm 0.01}$	**88.67**$_{\pm 0.52}$	-
EGFR-MET	RationaleRL	99.95$_{\pm 0.02}$	92.22$_{\pm 0.21}$	**100**$_{\pm 0.00}$	75.78$_{\pm 0.04}$	**99.81**$_{\pm 0.01}$
	REINVENT	99.15$_{\pm 1.55}$	94.82$_{\pm 0.14}$	99.80$_{\pm 0.05}$	89.59$_{\pm 0.04}$	70.47$_{\pm 0.12}$
	MARS	**100**$_{\pm 0.00}$	80.30$_{\pm 0.88}$	99.82$_{\pm 0.08}$	**91.89**$^{*}_{\pm 0.10}$	69.04$_{\pm 0.53}$
	CombiMOTS	**100**$_{\pm 0.00}$	**100**$_{\pm 0.00}$	99.81$_{\pm 1.62}$	90.75$_{\pm 0.40}$	-
PIK3CA-mTOR	RationaleRL	99.99$_{\pm 0.01}$	48.34$_{\pm 0.33}$	27.04$_{\pm 0.39}$	66.55$_{\pm 0.04}$	**99.98**$_{\pm 0.02}$
	REINVENT	99.47$_{\pm 0.08}$	98.39$_{\pm 0.09}$	99.96$_{\pm 0.01}$	87.71$^{*}_{\pm 0.04}$	97.60$_{\pm 0.20}$
	MARS	**100**$_{\pm 0.00}$	55.00$_{\pm 7.78}$	94.42$_{\pm 3.21}$	73.82$_{\pm 2.12}$	99.86$_{\pm 0.23}$
	CombiMOTS	**100**$_{\pm 0.00}$	**100**$_{\pm 0.00}$	**99.99**$_{\pm 0.01}$	**90.88**$_{\pm 0.36}$	-

Table 2. Average R2-distance to the utopia point normalized in range 0 to 2 (lower is better) and size of Pareto optimal sets.

Model/Task	GSK3β-JNK3	EGFR-MET	PIK3CA-mTOR
CombiMOTS	0.8075 **(174)**	**0.8419 (268)**	0.8143 **(210)**
MARS	**0.7830** (102)	0.8845 (142)	0.8144 (89)
RationaleRL	0.8307 (137)	0.9388 (148)	0.8852 (113)
REINVENT	0.8223 (93)	0.8518 (177)	**0.8035** (174)

of Tanimoto similarity thresholds on the search space size (Section 5.7.2).

5.7.1. Dual-Target-Aware Fragment Extraction

FGIB extends BRICS by breaking molecules into retrosynthetically interesting fragments, which is preferred over rule/frequency-based methods when combined with Enamine REAL Space. Some works adapt BRICS to needs with a fixed goal e.g. pBRICS (Vangala et al., 2023) for ADMET explanability, thus not suited for the dual-inhibition task where target proteins are user-defined. Our goal is to capture high-property fragments for any target property. To further justify our choice, we report results from 10k rollouts on the GSK3β-JNK3 task when replacing FGIB with naive BRICS and MiCaM (Geng et al., 2023), a connection-aware motif-mining approach to decompose known active compounds. Table 3 shows that FGIB successfully identified goal-specific fragments, resulting in a significantly higher rate of dual-active compounds. For other metrics (Appendix B), all methods exhibit similar performance, which is sound as FGIB only impacts the search space through the selection of initial blocks: MCTS objectives and convergence are "as good" but the attainable space contains more compounds likely exhibiting dual-activity potential.

5.7.2. Sensitivity of the Search Space Size

We investigate the use of various thresholds and report results from 10k rollouts on the GSK3β-JNK3 task for lower and higher values of 0.3, 0.5 & 0.6. Table 4 shows that (i) the threshold greatly affects the search space size due to the small nature of FGIB fragments and Enamine blocks. Tanimoto metric being fingerprint-based, changing a small motif is relatively impactful. (ii) For higher thresholds, the search space is too small to allow good exploration. There is a significant drop in number of dual-actives, and other metrics (Appendix B) suggest that thresholds above 0.6 yield a drop in diversity (88.67→78.73%) and consistency across molecular properties. (iii) For lower thresholds, CombiMOTS has "more room to explore" but finds worse tradeoffs across QED and less dual-actives. As more reactions are possible, Pareto fronts are larger and convergence to optimal solutions is slower (Appendix L). Optimal thresholds are specific to the target properties, but our experiments suggest that a search space of magnitude 10M yields better convergence within a reasonable budget (∼10k rollouts).

Table 3. Search space size and dual-actives candidates over all generations using FGIB, Naive BRICS and MiCaM. $\neq$ *refers to blocks not used in the specified method.*

Method	#Building Blocks	#Possible Products	#Dual-Actives
BRICS	4,430 (747 $\neq$ FGIB)	∼1.7M	1,815/12,559 (14.45%)
MiCaM	12,274 (8,428 $\neq$ FGIB)	∼26M	1,445/13,263 (10.90%)
FGIB (Base)	14,366 (10,683 $\neq$ BRICS, 10,520 $\neq$ MiCaM)	∼25M	3,662/15,423 (**23.74%**)

Table 4. Search space size and dual-actives candidates over all generations using different similarity threshold values.

Threshold	#Building Blocks	# Possible Products	# Dual-Actives
0 (Full Space)	139,493	>31B	(Not Performed)
0.3	54,811	∼478M	2,833/13,556 (20.90%)
0.4 (Base)	14,366	∼25M	3,662/15,423 (**23.74%**)
0.5	3,737	∼1.1M	1,380/11,632 (11.86%)
0.6	858	∼43k	317/9,433 (3.36%)

6. Case Studies

6.1. Visualization of potential dual-inhibitors for GSK3β-JNK3

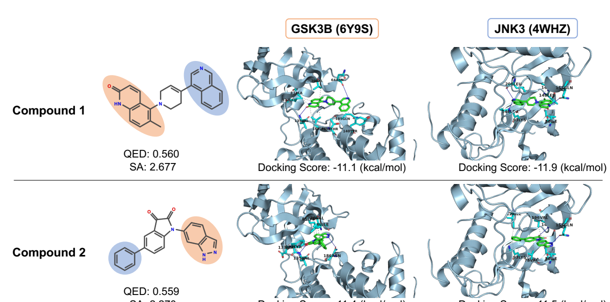

Figure 5. Examples of molecules generated by CombiMOTS on the GSK3β-JNK3 target pair. They both yield high drug-likeness and synthesizability metrics, and different (colored) parts of the compounds are positioned into each protein pocket.

We show in Figure 5 two example dual-target molecules

(Compound 1 and Compound 2) generated by CombiMOTS to inhibit both GSK3β and JNK3. Through *in silico* docking, each compound was predicted to occupy the ATP binding pocket of both targets, reproducing key interaction patterns highlighted in previous structural and biochemical studies (Lu et al., 2018; Cheng et al., 2022). Notably, hydrogen-bond interactions in the hinge region (e.g., *VAL135* in GSK3β; *MET149* in JNK3) and the engagement of hydrophobic pockets (*VAL70, LEU132* in GSK3β; *ILE70, VAL78* in JNK3) were consistently observed, indicating strong binding complementarity.

Compound 1 exhibited strong predicted affinity, with docking scores of –11.1 kcal/mol for GSK3β and –11.9 kcal/mol for JNK3. Hydrophobic interactions with residues such as *VAL70* (GSK3β) and *ILE70* (JNK3) stabilized its positioning in catalytic clefts. Hydrogen bonds with residues such as *ASN64* (GSK3β) and *LYS93* (JNK3) further reinforced dual-target binding, aligning with known hinge-directed inhibitor interactions (Quesada-Romero et al., 2014; Cheng et al., 2022).

Similarly, Compound 2 achieved favorable docking scores (–11.4 kcal/mol for GSK3β, –11.5 kcal/mol for JNK3). Key hydrogen bonds with *ASP133* (GSK3β) and *MET149* (JNK3), along with hydrophobic contacts involving *ILE62* (GSK3β) and *ILE70* (JNK3), support its effective active-site engagement (Quesada-Romero et al., 2014; Lu et al., 2018; Cheng et al., 2022). These interactions indicate robust dual-target binding for both compounds. The full list of identified interactions is found in Appendix G.

Beyond binding predictions, both compounds display QED values around 0.56 and SA scores under 3, suggesting a reasonable balance of drug-like properties and ease of synthesis. These findings highlight the promise of CombiMOTS-driven design in generating dual-target molecules that not only recapitulate essential interaction motifs but also meet fundamental criteria for drug discovery.

6.2. Toxicity Prediction

To further demonstrate the applicability of our method in drug discovery, we estimate the toxicity profile of molecules generated by CombiMOTS. Following Swanson et al. (2024), we train an ensemble of ten Chemprop models on the ClinTox dataset (Wu et al., 2018) using clinical toxicity binary labels. The dataset comprises 1,478 molecules, with 1,366 experimentally validated as non-toxic (0) and 112 identified as toxic (1).

Figure 6 compares the predicted toxicities of 10,000 CombiMOTS-generated molecules for the GSK3β-JNK3 task against ground truth compounds from ClinTox. While some generations tend to be toxic, we make two key observations: (i) The average predicted toxicity of 10,000 samples

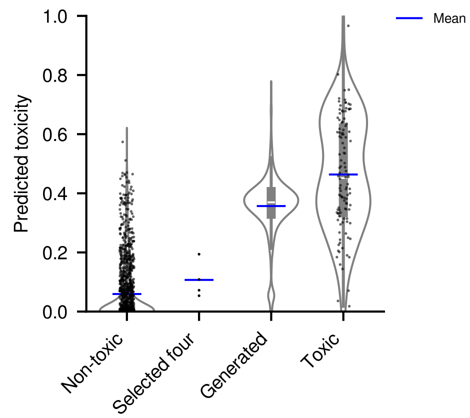

Figure 6. Toxicity predictions for generated molecules on the GSK3β-JNK3 target pair. 'Generated' refers to all 10,000 generated samples. 'Selected four' refers to the compounds discussed in 6.2.

remains below that of the 112 truly toxic compounds, with some generations predicted as potentially non-toxic. (ii) Four randomly selected samples with predicted toxicity below 0.2 exhibit high scores across all properties (Figure 19). Despite toxicity not being optimized, these four samples appear in the first two Pareto fronts, suggesting that Pareto-optimal solutions may hold promise for lead identification. We perform an additional experiment on toxicity optimization in Appendix I.2.

7. Conclusion

We proposed CombiMOTS, a novel approach to combinatorial multi-objective drug design by integrating PMCTS with fragment-based molecule generation. By leveraging dual-target-aware fragments available in industry and extending PMCTS to a combinatorial framework, our method effectively navigates the complex design space of dual-target drug discovery while ensuring synthesizability. Through comprehensive evaluations on multiple dual-target inhibitor tasks, we demonstrate that CombiMOTS consistently outperforms existing methods by generating diverse and interpretable molecules that achieve favorable trade-offs across key objectives. Our results highlight the potential of CombiMOTS as a valuable tool in rational drug design, paving the way for further advancements in multi-objective optimization for molecular generation.

Future directions could extend our framework to larger chemical libraries e.g., Freedom Space (Protopopov et al., 2024) or broader multi-objective applications. While Appendix I offers preliminary investigations of selective molecule generation, toxicity optimization and protein-protein interaction modulation, we envision future works to explore various strategies to expand the applicability of CombiMOTS in real-world scenarios.

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Impact Statement

We release CombiMOTS to be effective in real-world drug discovery tasks and useful to practitioners. Common risks involve using this tool to design harmful compounds or having complete faith in machine-learned predicted properties with no *post-hoc* validation. For instance, metrics as validity, uniqueness, novelty and diversity can easily be diverted through small modifications on an original SMILES string (Renz et al., 2019). Though such challenges remain common in other domains such as language and image, we encourage users to work closely with application-driven experts such as medicinal chemists or health institutes.

References

Agu, P. C., Afiukwa, C. A., Orji, O., Ezeh, E., Ofoke, I., Ogbu, C., Ugwuja, E. I., and Aja, P. Molecular docking as a tool for the discovery of molecular targets of nutraceuticals in diseases management. *Scientific reports*, 13(1):13398, 2023.

Angelo, J. S., Guedes, I. A., Barbosa, H. J., and Dardenne, L. E. Multi-and many-objective optimization: present and future in de novo drug design. *Frontiers in Chemistry*, 11:1288626, 2023.

Blaschke, T., Arús-Pous, J., Chen, H., Margreitter, C., Tyrchan, C., Engkvist, O., Papadopoulos, K., and Patronov, A. Reinvent 2.0: an ai tool for de novo drug design. *Journal of chemical information and modeling*, 60(12): 5918–5922, 2020.

Chen, S. and Jung, Y. Estimating the synthetic accessibility of molecules with building block and reaction-aware sascore. *Journal of Cheminformatics*, 16(1):83, 2024.

Chen, S., Xie, J., Ye, R., Xu, D. D., and Yang, Y. Structure-aware dual-target drug design through collaborative learning of pharmacophore combination and molecular simulation. *Chemical Science*, 15(27):10366–10380, 2024.

Chen, W. and Liu, L. Pareto monte carlo tree search for multi-objective informative planning. *arXiv preprint arXiv:2111.01825*, 2021.

Cheng, C., Liu, M., Gao, X., Wu, D., Pu, M., Ma, J., Quinn, R. J., Xiao, Z., and Liu, Z. Identifying new ligands for jnk3 by fluorescence thermal shift assays and native mass spectrometry. *Acs Omega*, 7(16):13925–13931, 2022.

Chenthamarakshan, V., Das, P., Hoffman, S., Strobelt, H., Padhi, I., Lim, K. W., Hoover, B., Manica, M., Born, J., Laino, T., et al. Cogmol: Target-specific and selective drug design for covid-19 using deep generative models. *Advances in Neural Information Processing Systems*, 33: 4320–4332, 2020.

Cieplinski, T., Danel, T., Podlewska, S., and Jastrzebski, S. We should at least be able to design molecules that dock well. *arXiv preprint arXiv:2006.16955*, 2020.

Constantin, T. A., Varela-Carver, A., Greenland, K. K., de Almeida, G. S., Olden, E., Penfold, L., Ang, S., Ormrod, A., Leach, D. A., Lai, C.-F., et al. The cdk7 inhibitor ct7001 (samuraciclib) targets proliferation pathways to inhibit advanced prostate cancer. *British Journal of Cancer*, 128(12):2326–2337, 2023.

Degen, J., Wegscheid-Gerlach, C., Zaliani, A., and Rarey, M. On the art of compiling and using'drug-like'chemical fragment spaces. *ChemMedChem*, 3(10):1503, 2008.

Du, H., Jiang, D., Zhang, O., Wu, Z., Gao, J., Zhang, X., Wang, X., Deng, Y., Kang, Y., Li, D., et al. A flexible data-free framework for structure-based de novo drug design with reinforcement learning. *Chemical Science*, 14(43):12166–12181, 2023.

El-Sharkawi, A. Pareto multi objective optimization. In *Proceedings of the 13th international conference on, intelligent systems application to power systems*, pp. 84–91. IEEE, 2005.

Ertl, P. and Schuffenhauer, A. Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *Journal of cheminformatics*, 1:1–11, 2009.

Fang, M.-M., Cheng, J.-T., Chen, Y.-Q., Lin, X.-C., Su, J.-W., Wu, Y.-L., Chen, H.-J., and Yang, J.-J. Molecular features and clinical outcomes of egfr-mutated, met-amplified non-small-cell lung cancer after resistance to dual-targeted therapy. *Therapeutic Advances in Medical Oncology*, 16:17588359241234504, 2024.

Feldmann, C., Philipps, M., and Bajorath, J. Explainable machine learning predictions of dual-target compounds reveal characteristic structural features. *Scientific Reports*, 11(1):21594, 2021.

Fisher, R. P. Secrets of a double agent: Cdk7 in cell-cycle control and transcription. *Journal of cell science*, 118 (22):5171–5180, 2005.

Foucquier, J. and Guedj, M. Analysis of drug combinations: current methodological landscape. *Pharmacology research & perspectives*, 3(3):e00149, 2015.

Fromer, J. C. and Coley, C. W. Computer-aided multi-objective optimization in small molecule discovery. *Patterns*, 4(2):100678, 2023. ISSN 2666-3899.

Fruman, D. A. and Rommel, C. Pi3k and cancer: lessons, challenges and opportunities. *Nature reviews Drug discovery*, 13(2):140–156, 2014.

Gao, W. and Coley, C. W. The synthesizability of molecules proposed by generative models. *Journal of chemical information and modeling*, 60(12):5714–5723, 2020.

Gaulton, A., Bellis, L. J., Bento, A. P., Chambers, J., Davies, M., Hersey, A., Light, Y., McGlinchey, S., Michalovich, D., Al-Lazikani, B., et al. Chembl: a large-scale bioactivity database for drug discovery. *Nucleic acids research*, 40(D1):D1100–D1107, 2012.

Geng, Z., Xie, S., Xia, Y., Wu, L., Qin, T., Wang, J., Zhang, Y., Wu, F., and Liu, T.-Y. De novo molecular generation via connection-aware motif mining. *arXiv preprint arXiv:2302.01129*, 2023.

Genheden, S., Thakkar, A., Chadimová, V., Reymond, J.-L., Engkvist, O., and Bjerrum, E. Aizynthfinder: a fast, robust and flexible open-source software for retrosynthetic planning. *Journal of cheminformatics*, 12(1):70, 2020.

Ghose, A. K., Viswanadhan, V. N., and Wendoloski, J. J. A knowledge-based approach in designing combinatorial or medicinal chemistry libraries for drug discovery. 1. a qualitative and quantitative characterization of known drug databases. *Journal of combinatorial chemistry*, 1(1): 55–68, 1999.

Gong, H., Liu, Q., Wu, S., and Wang, L. Text-guided molecule generation with diffusion language model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 109–117, 2024.

Grygorenko, O. O., Radchenko, D. S., Dziuba, I., Chuprina, A., Gubina, K. E., and Moroz, Y. S. Generating multi-billion chemical space of readily accessible screening compounds. *Iscience*, 23(11), 2020.

Guimaraes, G. L., Sanchez-Lengeling, B., Outeiral, C., Farias, P. L. C., and Aspuru-Guzik, A. Objective-reinforced generative adversarial networks (organ) for sequence generation models. *arXiv preprint arXiv:1705.10843*, 2017.

Hart, S., Novotny-Diermayr, V., Goh, K. C., Williams, M., Tan, Y. C., Ong, L. C., Cheong, A., Ng, B. K., Amalini, C., Madan, B., et al. Vs-5584, a novel and highly selective pi3k/mtor kinase inhibitor for the treatment of cancer. *Molecular cancer therapeutics*, 12(2):151–161, 2013.

Hopkins, A. L., Mason, J. S., and Overington, J. P. Can we rationally design promiscuous drugs? *Current opinion in structural biology*, 16(1):127–136, 2006.

Huang, F., Han, X., Xiao, X., and Zhou, J. Covalent warheads targeting cysteine residue: The promising approach in drug development. *Molecules*, 27(22):7728, 2022.

Jin, W., Barzilay, R., and Jaakkola, T. Junction tree variational autoencoder for molecular graph generation. In *International conference on machine learning*, pp. 2323–2332. PMLR, 2018.

Jin, W., Barzilay, R., and Jaakkola, T. Multi-objective molecule generation using interpretable substructures. In *International conference on machine learning*, pp. 4849–4859. PMLR, 2020.

Kalgutkar, A. S. Designing around structural alerts in drug discovery. *Journal of Medicinal Chemistry*, 63(12):6276–6302, 2019.

Kim, S., Thiessen, P. A., Bolton, E. E., Chen, J., Fu, G., Gindulyte, A., Han, L., He, J., He, S., Shoemaker, B. A., et al. Pubchem substance and compound databases. *Nucleic acids research*, 44(D1):D1202–D1213, 2016.

Koch, P., Gehringer, M., and Laufer, S. A. Inhibitors of c-jun n-terminal kinases: an update. *Journal of medicinal chemistry*, 58(1):72–95, 2015.

Kocsis, L. and Szepesvári, C. Bandit based monte-carlo planning. In *European conference on machine learning*, pp. 282–293. Springer, 2006.

Kollat, J. B., Reed, P. M., and Kasprzyk, J. R. A new epsilon-dominance hierarchical bayesian optimization algorithm for large multiobjective monitoring network design problems. *Advances in Water Resources*, 31(5): 828–845, 2008.

Kong, X., Huang, W., Tan, Z., and Liu, Y. Molecule generation by principal subgraph mining and assembling. *Advances in Neural Information Processing Systems*, 35: 2550–2563, 2022.

Kummar, S., Chen, H. X., Wright, J., Holbeck, S., Millin, M. D., Tomaszewski, J., Zweibel, J., Collins, J., and Doroshow, J. H. Utilizing targeted cancer therapeutic agents in combination: novel approaches and urgent requirements. *Nature reviews Drug discovery*, 9(11):843–856, 2010.

Kusanda, N., Tom, G., Hickman, R., Nigam, A., Jorner, K., and Aspuru-Guzik, A. Assessing multi-objective optimization of molecules with genetic algorithms against relevant baselines. In *AI for Accelerated Materials Design NeurIPS 2022 Workshop*, 2022.

Kwiatkowski, N., Zhang, T., Rahl, P. B., Abraham, B. J., Reddy, J., Ficarro, S. B., Dastur, A., Amzallag, A., Ramaswamy, S., Tesar, B., et al. Targeting transcription regulation in cancer with a covalent cdk7 inhibitor. *Nature*, 511(7511):616–620, 2014.

Lai, H., Kannas, C., Hassen, A. K., Granqvist, E., Westerlund, A. M., Clevert, D.-A., Preuss, M., and Genheden, S. Multi-objective synthesis planning by means of monte carlo tree search. *Artificial Intelligence in the Life Sciences*, 7:100130, 2025.

Layman, R. M., Han, H. S., Rugo, H. S., Stringer-Reasor, E. M., Specht, J. M., Dees, E. C., Kabos, P., Suzuki, S., Mutka, S. C., Sullivan, B. F., et al. Gedatolisib in combination with palbociclib and endocrine therapy in women with hormone receptor-positive, her2-negative advanced breast cancer: results from the dose expansion groups of an open-label, phase 1b study. *The Lancet Oncology*, 25(4):474–487, 2024.

Lee, S., Lee, S., Kawaguchi, K., and Hwang, S. J. Drug discovery with dynamic goal-aware fragments. *arXiv preprint arXiv:2310.00841*, 2023.

Li, Y., Zhang, L., and Liu, Z. Multi-objective de novo drug design with conditional graph generative model. *Journal of cheminformatics*, 10:1–24, 2018.

Li, Y. H., Wang, P. P., Li, X. X., Yu, C. Y., Yang, H., Zhou, J., Xue, W. W., Tan, J., and Zhu, F. The human kinome targeted by fda approved multi-target drugs and combination products: A comparative study from the drug-target interaction network perspective. *PloS one*, 11(11):e0165737, 2016.

Lin, H.-H., Zhang, L.-L., Yan, R., Lu, J.-J., and Hu, Y. Network analysis of drug–target interactions: a study on fda-approved new molecular entities between 2000 to 2015. *Scientific reports*, 7(1):12230, 2017.

Liu, R., Li, X., and Lam, K. S. Combinatorial chemistry in drug discovery. *Current opinion in chemical biology*, 38: 117–126, 2017.

Lu, K., Wang, X., Chen, Y., Liang, D., Luo, H., Long, L., Hu, Z., and Bao, J. Identification of two potential glycogen synthase kinase 3β inhibitors for the treatment of osteosarcoma. *Acta Biochimica et Biophysica Sinica*, 50(5):456–464, 2018.

Luc, D. T. Pareto optimality. *Pareto optimality, game theory and equilibria*, pp. 481–515, 2008.

Luukkonen, S., van den Maagdenberg, H. W., Emmerich, M. T., and van Westen, G. J. Artificial intelligence in multi-objective drug design. *Current Opinion in Structural Biology*, 79:102537, 2023. ISSN 0959-440X.

Maziarz, K., Jackson-Flux, H., Cameron, P., Sirockin, F., Schneider, N., Stiefl, N., Segler, M., and Brockschmidt, M. Learning to extend molecular scaffolds with structural motifs. *arXiv preprint arXiv:2103.03864*, 2021.

McCubrey, J. A., Davis, N. M., Abrams, S. L., Montalto, G., Cervello, M., Basecke, J., Libra, M., Nicoletti, F., Cocco, L., Martelli, A. M., et al. Diverse roles of gsk-3: Tumor promoter–tumor suppressor, target in cancer therapy. *Advances in biological regulation*, 54:176–196, 2014.

Medina-Franco, J. L., Giulianotti, M. A., Welmaker, G. S., and Houghten, R. A. Shifting from the single to the multitarget paradigm in drug discovery. *Drug discovery today*, 18(9-10):495–501, 2013.

Morphy, J. R. and Harris, C. J. *Designing multi-target drugs*, volume 21. Royal Society of Chemistry, 2012.

O'Boyle, N. M., Banck, M., James, C. A., Morley, C., Vandermeersch, T., and Hutchison, G. R. Open babel: An open chemical toolbox. *Journal of cheminformatics*, 3: 1–14, 2011.

Olivecrona, M., Blaschke, T., Engkvist, O., and Chen, H. Molecular de-novo design through deep reinforcement learning. *Journal of cheminformatics*, 9:1–14, 2017.

Olson, C. M., Liang, Y., Leggett, A., Park, W. D., Li, L., Mills, C. E., Elsarrag, S. Z., Ficarro, S. B., Zhang, T., Düster, R., et al. Development of a selective cdk7 covalent inhibitor reveals predominant cell-cycle phenotype. *Cell chemical biology*, 26(6):792–803, 2019.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. Scikit-learn: Machine learning in python. *the Journal of machine Learning research*, 12:2825–2830, 2011.

Polishchuk, P. G., Madzhidov, T. I., and Varnek, A. Estimation of the size of drug-like chemical space based on gdb-17 data. *Journal of computer-aided molecular design*, 27:675–679, 2013.

Popova, M., Isayev, O., and Tropsha, A. Deep reinforcement learning for de novo drug design. *Science advances*, 4(7): eaap7885, 2018.

Powers, A. S., Yu, H. H., Suriana, P., Koodli, R. V., Lu, T., Paggi, J. M., and Dror, R. O. Geometric deep learning for structure-based ligand design. *ACS Central Science*, 9(12):2257–2267, 2023.

Protopopov, M. V., Tararina, V. V., Bonachera, F., Dzyuba, I. M., Kapeliukha, A., Hlotov, S., Chuk, O., Marcou, G., Klimchuk, O., Horvath, D., et al. The freedom space–a new set of commercially available molecules for hit discovery. *Molecular Informatics*, 43(12):e202400114, 2024.

Quesada-Romero, L., Mena-Ulecia, K., Tiznado, W., and Caballero, J. Insights into the interactions between maleimide derivates and gsk3β combining molecular docking and qsar. *PLoS One*, 9(7):e102212, 2014.

Raghavendra, N. M., Pingili, D., Kadasi, S., Mettu, A., and Prasad, S. Dual or multi-targeting inhibitors: The next generation anticancer agents. *European journal of medicinal chemistry*, 143:1277–1300, 2018.

Ren, X., Li, K., Zhang, Y., Zou, C., and Su, M. Case report: Response to savolitinib/egfr-tki combination in nsclc patients harboring concurrent primary met amplification/overexpression and egfr mutation. *Frontiers in Oncology*, 14:1297156, 2024.

Renz, P., Van Rompaey, D., Wegner, J. K., Hochreiter, S., and Klambauer, G. On failure modes in molecule generation and optimization. *Drug Discovery Today: Technologies*, 32:55–63, 2019.

Rodrik-Outmezguine, V. S., Okaniwa, M., Yao, Z., Novotny, C. J., McWhirter, C., Banaji, A., Won, H., Wong, W., Berger, M., de Stanchina, E., et al. Overcoming mtor resistance mutations with a new-generation mtor inhibitor. *Nature*, 534(7606):272–276, 2016.

Sava, G. P., Fan, H., Coombes, R. C., Buluwela, L., and Ali, S. Cdk7 inhibitors as anticancer drugs. *Cancer and Metastasis Reviews*, 39:805–823, 2020.

Segler, M. H. and Waller, M. P. Neural-symbolic machine learning for retrosynthesis and reaction prediction. *Chemistry–A European Journal*, 23(25):5966–5971, 2017.

Segler, M. H., Preuss, M., and Waller, M. P. Planning chemical syntheses with deep neural networks and symbolic ai. *Nature*, 555(7698):604–610, 2018.

Stanley, M. and Segler, M. Fake it until you make it? generative de novo design and virtual screening of synthesizable molecules. *Current Opinion in Structural Biology*, 82: 102658, 2023.

Sun, H., Li, J., Zhang, Y., Lin, S., Chen, J., Tan, H., Wang, R., Mao, X., Zhao, J., Li, R., et al. Target-specific design of drug-like ppi inhibitors via hot-spot-guided generative deep learning. *bioRxiv*, pp. 2024–10, 2024.

Sun, J., Jeliazkova, N., Chupakhin, V., Golib-Dzib, J.-F., Engkvist, O., Carlsson, L., Wegner, J., Ceulemans, H., Georgiev, I., Jeliazkov, V., et al. Excape-db: an integrated large scale dataset facilitating big data analysis in chemogenomics. *Journal of cheminformatics*, 9:1–9, 2017.

Sun, M., Xing, J., Meng, H., Wang, H., Chen, B., and Zhou, J. Molsearch: search-based multi-objective molecular generation and property optimization. In *Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining*, pp. 4724–4732, 2022.

Suzuki, T., Ma, D., Yasuo, N., and Sekijima, M. Mothra: Multiobjective de novo molecular generation using monte carlo tree search. *Journal of Chemical Information and Modeling*, 64(19):7291–7302, 2024.

Swanson, K., Liu, G., Catacutan, D. B., Arnold, A., Zou, J., and Stokes, J. M. Generative ai for designing and validating easily synthesizable and structurally novel antibiotics. *Nature Machine Intelligence*, 6(3):338–353, 2024.

Tan, Y., Dai, L., Huang, W., Guo, Y., Zheng, S., Lei, J., Chen, H., and Yang, Y. Drlinker: Deep reinforcement learning for optimization in fragment linking design. *Journal of Chemical Information and Modeling*, 62(23):5907–5917, 2022.

Tang, S., Ding, J., Zhu, X., Wang, Z., Zhao, H., and Wu, J. Vina-gpu 2.1: towards further optimizing docking speed and precision of autodock vina and its derivatives. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2024.

Thakkar, A., Chadimová, V., Bjerrum, E. J., Engkvist, O., and Reymond, J.-L. Retrosynthetic accessibility score (rascore)–rapid machine learned synthesizability classification from ai driven retrosynthetic planning. *Chemical science*, 12(9):3339–3349, 2021.

Vangala, S. R., Krishnan, S. R., Bung, N., Srinivasan, R., and Roy, A. pbrics: a novel fragmentation method for explainable property prediction of drug-like small molecules. *Journal of Chemical Information and Modeling*, 63(16):5066–5076, 2023.

Vinkers, H. M., de Jonge, M. R., Daeyaert, F. F., Heeres, J., Koymans, L. M., van Lenthe, J. H., Lewi, P. J., Timmerman, H., Van Aken, K., and Janssen, P. A. Synopsis:

synthesize and optimize system in silico. *Journal of medicinal chemistry*, 46(13):2765–2773, 2003.

Wahab, A. and Gershoni-Poranne, R. Compas-3: a dataset of peri-condensed polybenzenoid hydrocarbons. *Physical Chemistry Chemical Physics*, 26(21):15344–15357, 2024.

Wang, J., Mao, J., Li, C., Xiang, H., Wang, X., Wang, S., Wang, Z., Chen, Y., Li, Y., No, K. T., et al. Interface-aware molecular generative framework for protein–protein interaction modulators. *Journal of Cheminformatics*, 16(1):142, 2024.

Wu, T., Ren, H., Li, P., and Leskovec, J. Graph information bottleneck. *Advances in Neural Information Processing Systems*, 33:20437–20448, 2020.

Wu, Z., Ramsundar, B., Feinberg, E. N., Gomes, J., Geniesse, C., Pappu, A. S., Leswing, K., and Pande, V. Moleculenet: a benchmark for molecular machine learning. *Chemical science*, 9(2):513–530, 2018.

Xie, Y., Shi, C., Zhou, H., Yang, Y., Zhang, W., Yu, Y., and Li, L. Mars: Markov molecular sampling for multi-objective drug discovery. *arXiv preprint arXiv:2103.10432*, 2021.

Xu, X., Zhou, J., Zhu, C., Zhan, Q., Li, Z., Zhang, R., Wang, Y., Liao, X., and Gao, X. Optimization of binding affinities in chemical space with generative pre-trained transformer and deep reinforcement learning. *F1000Research*, 12:757, 2024.

Xue, Z., Sun, C., Zheng, W., Lv, J., and Liu, X. Targetsa: adaptive simulated annealing for target-specific drug design. *Bioinformatics*, 41(1):btae730, 2025.

Yang, S., Hwang, D., Lee, S., Ryu, S., and Hwang, S. J. Hit and lead discovery with explorative rl and fragment-based molecule generation. *Advances in Neural Information Processing Systems*, 34:7924–7936, 2021.

Yang, Y., Chen, G., Li, J., Li, J., Zhang, O., Zhang, X., Li, L., Hao, J., Wang, E., and Heng, P.-A. Enabling target-aware molecule generation to follow multi objectives with pareto mcts. *Communications Biology*, 7(1):1074, 2024a.

Yang, Y., Mou, Y., Wan, L.-X., Zhu, S., Wang, G., Gao, H., and Liu, B. Rethinking therapeutic strategies of dual-target drugs: An update on pharmacological small-molecule compounds in cancer. *Medicinal Research Reviews*, 2024b.

Ye, J., Wu, J., and Liu, B. Therapeutic strategies of dual-target small molecules to overcome drug resistance in cancer therapy. *Biochimica et Biophysica Acta (BBA)-Reviews on Cancer*, 1878(3):188866, 2023.

Zeng, X., Wang, F., Luo, Y., Kang, S.-g., Tang, J., Lightstone, F. C., Fang, E. F., Cornell, W., Nussinov, R., and Cheng, F. Deep generative molecular design reshapes drug discovery. *Cell Reports Medicine*, 3(12), 2022.

Zhou, X., Guan, J., Zhang, Y., Peng, X., Wang, L., and Ma, J. Reprogramming pretrained target-specific diffusion models for dual-target drug design. *arXiv preprint arXiv:2410.20688*, 2024.